

GEOGRAPHY

INDIAN GEOGRAPHY



MINERALS & INDUSTRIES

MODULE - 7

**IRON AND STEEL
INDUSTRY**

GS PAPER 1 SYLLABUS

- *Distribution of key natural resources;*
- *Factors responsible for the location of primary, secondary, and tertiary sector industries in various parts of the world (including India)*

IRON AND STEEL INDUSTRY

AN INTRODUCTION

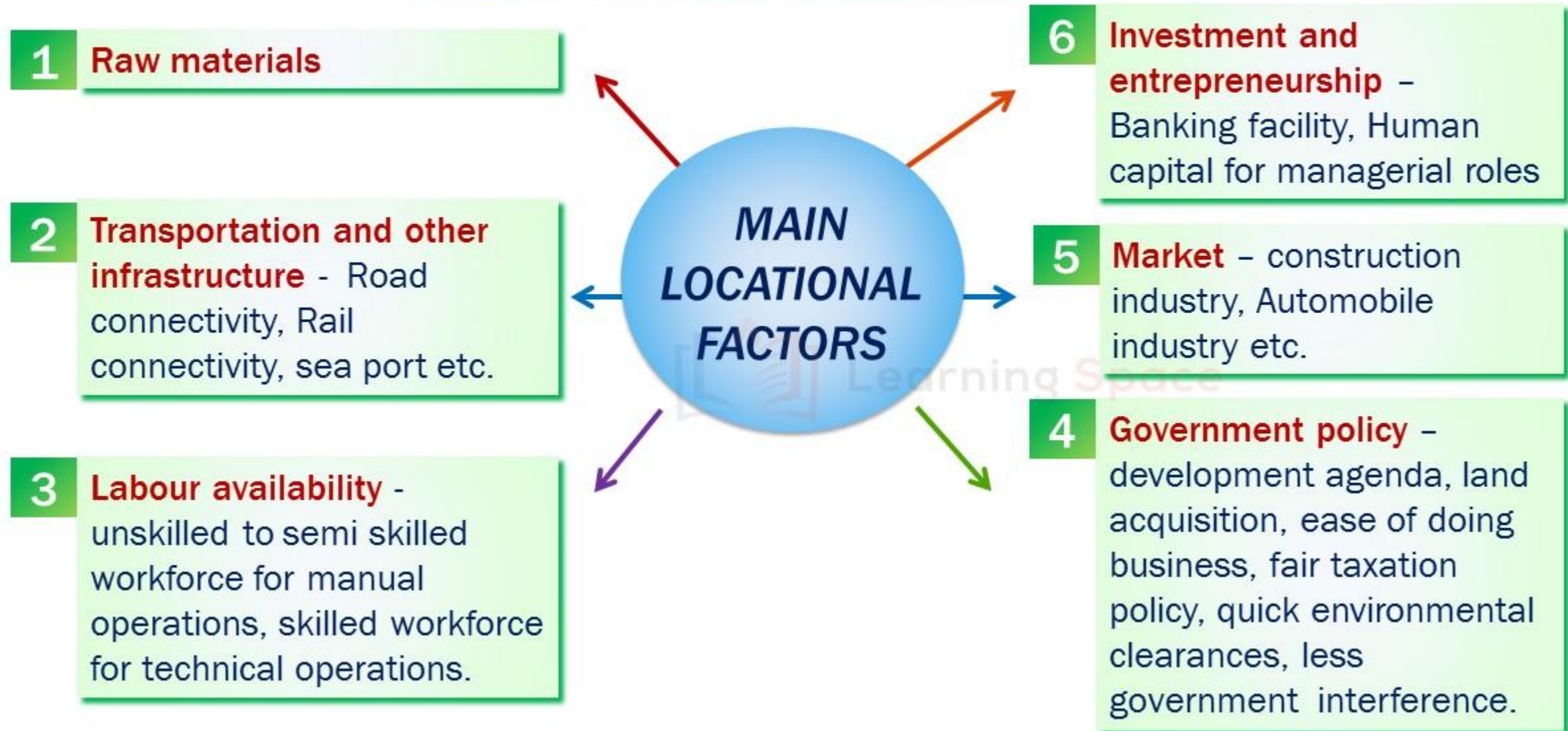
SPECIAL PROPERTIES INDUCED BY ADDITION OF CERTAIN CHEMICALS:

- **Chromium** is added to make stainless steel.
- **Cobalt** is added to induce magnetic properties (used in turbine making); high grade steel; ferrite strengthener.
- **Vanadium** is used to induce elasticity property in steel (manufacturing of springs).
- **Tungsten** and **manganese** are added to increase the strength of the steel.

- 1 The development of Iron and Steel industry opened the doors to rapid industrial development in India.
- 2 Almost all sectors of the Indian industry depend heavily on the Iron and Steel industry for their basic infrastructure.
- 3 Raw materials of the industry-- **Iron ore, coking coal, lime stone** etc.
- 4 ***All these raw materials are gross in nature*** (weight losing). Therefore the best location for the Iron and Steel plant is ***near the source of raw material.***

IRON AND STEEL INDUSTRY

FACTORS THAT INFLUENCE THE LOCATION OF IRON AND STEEL INDUSTRY



IRON AND STEEL INDUSTRY

PROCESS INVOLVED IN IRON AND STEEL INDUSTRY

To understand about the factor that influence the location of Iron and Steel industry, we have to understand about Iron ore smelting process.



WHAT IS SMELTING?

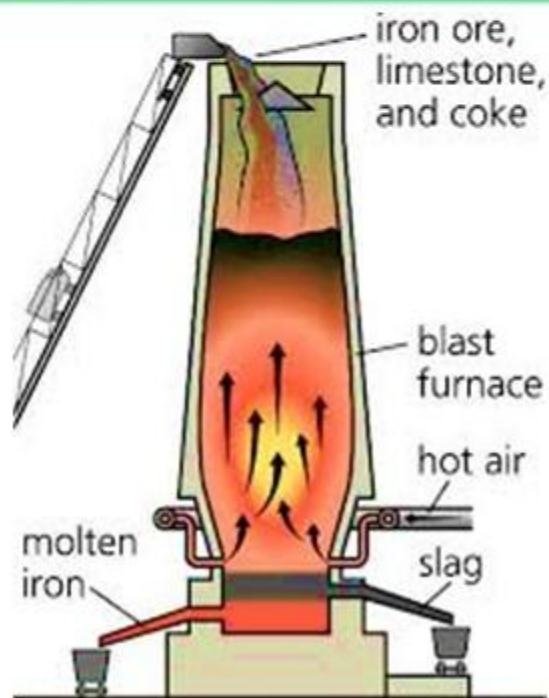
Smelting is a process of converting ore to metal by removing impurities.



IRON AND STEEL INDUSTRY

WHAT EXACTLY HAPPENS IN THE SMELTING PROCESS?

- 1 There are many impurities present in Iron ore. Some are beneficial when present in small quantities, while others are harmful no matter what the proportion is.



- 2 So, the unwanted impurities must be removed and this is done by smelting of iron-ore in a blast-furnace.
- 3 After **extraction** of the crude mineral is done, **concentration** of the Iron-ore takes place, this is done through the process of **magnetic separation**, **gravitational filtration**, **ore washing** etc.
- 4 After concentration process, in a blast-furnace fuel (coke), **Iron-ore** and **flux** (limestone) are continuously supplied through the top of the furnace.

IRON AND STEEL INDUSTRY

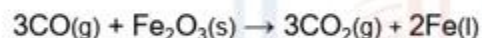
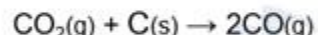
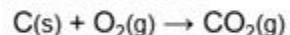
WHAT EXACTLY HAPPENS IN THE SMELTING PROCESS?

Extraction of Iron

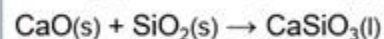
Iron ore (Fe_2O_3)
Coke (C)
Limestone (CaCO_3)

Waste gases
 CO_2 , CO

Main reactions:



Major impurity in iron ore is sand $\text{SiO}_2\text{(s)}$



Slag is CaSiO_3 and other impurities.

Hot air blast.

Hot air blast.

Slag floats above molten iron.

Molten iron sinks to bottom.

Slag tapped off separately.

Molten iron tapped off.

5 A hot blast of air is blown in to the lower section.

6 In a blast-furnace, iron oxides are converted into liquid iron.

Iron-ore + Coke + Limestone → Pig Iron + Slag + Carbon-dioxide

7 The molten iron which we get as output from the blast furnace is called pig iron.

IRON AND STEEL INDUSTRY

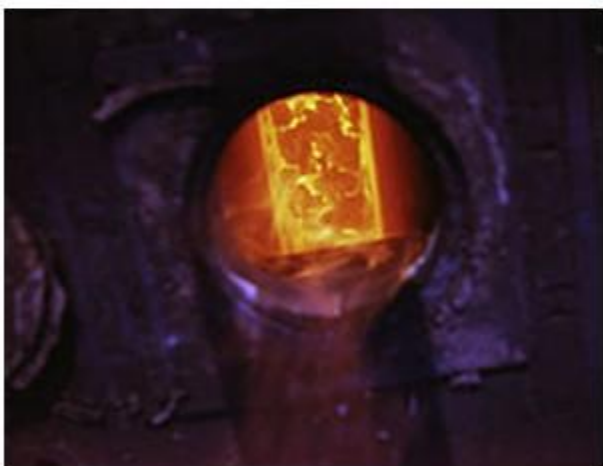
WHAT EXACTLY HAPPENS IN THE SMELTING PROCESS?



- 8 This molten liquid iron is then converted into steel in a Bessemer convertor.
- 9 In a Bessemer convertor molten iron, manganese, nickel and cobalt is added to reduce the brittleness of the iron and to induce strength to the steel which will be the final output product.

IRON AND STEEL INDUSTRY

WHY COKE AND NOT COAL IN SMELTING PROCESS?



- 1 To separate impurities iron ore needs to be melted.

- 2 The coke is actually the fuel that melts iron.

- 3 Coal has many impurities and the most dangerous one is sulphur.

- 4 Through a process called destructive distillation coke is produced from coal.

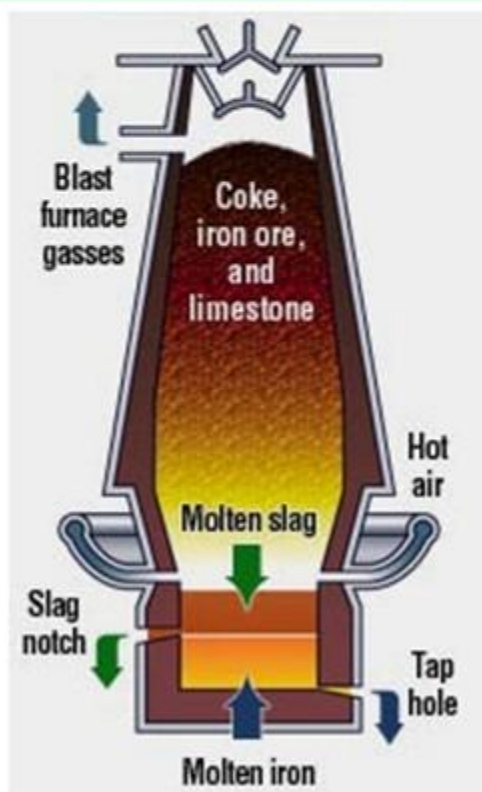
- 5 Coke is a fuel with very low impurities and a high carbon content.

- 6 The cooked coal called coke contains 92-93% carbon, some ash and sulphur but compared to raw coal it is very strong.

IRON AND STEEL INDUSTRY

ROLE OF LIMESTONE

- 1 Limestone acts as a **flux**.
- 2 Flux is a substance **which is mixed with solid to lower the melting point especially in the process of smelting.**



- 3 Such a flux combines with impurities to form **slag** which is an important raw material used in **cement industry**.
- 4 The sulphur which is present in the iron-ore gets removed as **slag** in the form of **calcium sulphide**.
- 5 The slag is also formed from any remaining silica (SiO_2), alumina (Al_2O_3) etc.
- 6 The liquid slag then trickles to a lot of furnace where it **floats on the top of liquid iron** since it is less dense, the pig iron settles at the bottom which can be conveniently removed.

IRON AND STEEL INDUSTRY

COMMONLY FOUND IMPURITIES IN IRON-ORE

Let us know discuss about some of the impurities which is found in Iron-ore.

1 SILICON

- 1 If present in small quantities it is actually beneficial as it rises the strength & hardness of steel. But the presence of its oxides is detrimental to iron.

2 SULPHUR

- 1 This a very harmful element.
- 2 It form a iron sulphide which is a brittle component.
- 3 It greatly **reduces the strength of steel.**

3 LEAD

- 1 It is added to all classes of steel to improve the alloy-ability property of the steel.
- 2 This impurity improves tool life.
- 3 Therefore, small quantities is actually good.

IRON AND STEEL INDUSTRY

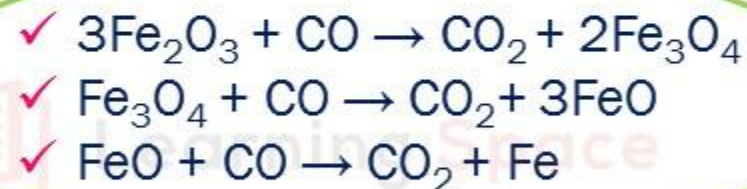
COMMONLY FOUND IMPURITIES IN IRON-ORE

4

OXYGEN

- 1 It has a bad influence on the property of steel.
- 2 Oxides makes iron and steel weak, therefore, it is very bad.
- 3 To remove oxygen, a reduction process is done by a series of chemical reaction.

- 4 Carbon monoxide is produced by burning coke.



- 5 Here, in these reactions you can understand that carbon monoxide which is produced by burning the coke acts as a **reducing agent**.

IRON AND STEEL INDUSTRY

WHAT DO YOU MEAN BY BENEFICIATION PROCESS?



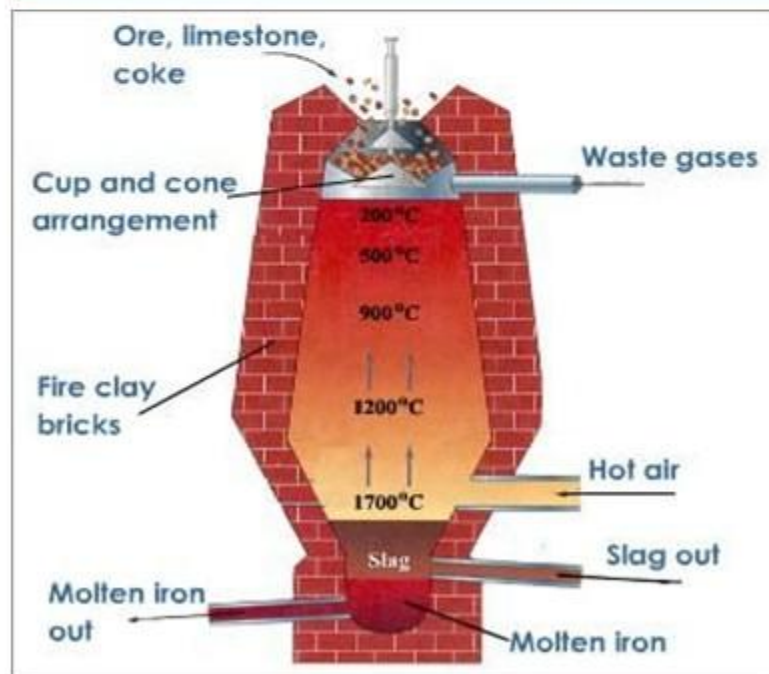
- 1 Beneficiation is the process by which, the **concentration of iron is improved**.
- 2 Certain iron-ore which is very rich in the iron content can be used as an input directly into a blast-furnace without any further processing.
- 3 Iron rich ore like hematite or magnetite contains iron in the ranges of 60-70%.
- 4 **Iron-ore that contains lower iron content must be processed or beneficiated to increase its content.**

IRON AND STEEL INDUSTRY

WHAT DO YOU MEAN BY BENEFICIATION PROCESS?

INPUTS INTO BLAST-FURNACE

- Ore = Iron-ore
- Fuel = Coke
- Flux = Limestone



OUTPUTS UPON SUCCESSIVE PROCESSING

- **Pig iron** – the intermediate product of smelting iron-ore.
- **Cast iron** – It has carbon content greater than 2% and is used in automotive industrial parts.
- **Stainless steel**
 - ✓ An alloy with a minimum of 10% chromium
 - ✓ It also contains nickel, manganese, molybdenum etc.
 - ✓ Stainless steel does not readily corrode, rust as ordinary steel.

IRON-ORE DISTRIBUTION ACROSS THE WORLD

IRON-ORE IN CHINA

- 1 Manchuria region of China is rich in with minerals.
- 2 Japan and Russia tried hard to capture this region, many wars were fought around it.
- 3 Shanghai Steel Industry is one of the very famous steel industry in the world.
- 4 However, only low grade iron-ore is found all across the China. So, China depends on imported ore from Australia, Brazil and Russia.
- 5 It produces steel domestically and it leads the world in steel exports.
- 6 In fact, dumping of steel from China is one of the important problem that Indian steel industry faces.
- 8 Therefore, companies are facing a stiff competition from the cheap Chinese goods and they keep demanding for a higher minimum import duty for steel.
- 9 Though we have good deposits of iron-ore, coking coal needs to be imported from either Australia or Indonesia.



IRON-ORE DISTRIBUTION ACROSS THE WORLD

IRON-ORE IN EUROPE



Mercedes-Benz



1 Iron-ore in Europe – Ruhr valley, Alsace – Lorraine coal fields, South Wales, Krivoy- Rog iron reserves of Ukraine

2 Flourishing automobile industry proves the fact that, there is enough iron and steel industry in the European side to sustain the automobile industry.

3 For example, you have automobile companies like Volkswagen, Benz, Audi, BMW which proves the efficiency of the automobile industries in Europe.

IRON-ORE DISTRIBUTION ACROSS THE WORLD

IRON-ORE IN NORTH AMERICA



1

The Great Lakes have contributed very much to the economic development of the United States and Canada.

2

Great Lakes Region, states of Pennsylvania, west Virginia, Alabama, Texas have good quality iron ore.

3

About 70% of all the iron ore of the United States is mined near Lake Superior.

4

Much of the coal produced in the Appalachian coal fields are brought to the Great Lakes because of which the iron and steel industry is well developed here.

5

In the Great Lakes Region, you have the world famous Detroit automobile industry which proves the worth of iron and steel industry of North America.

IRON-ORE DISTRIBUTION ACROSS THE WORLD

IRON-ORE IN SOUTH AMERICA



- 1 **Minas Geraais** is one of the important centre in Brazil which has good iron-ore.
- 2 Most of the ore from Brazil is exported to China and Japan.
- 3 **Carajas mines** is the largest iron-ore mine in the world.
- 4 Largest deposits of high-grade iron-ore are found here.

IRON-ORE DISTRIBUTION ACROSS THE WORLD



IRON-ORE IN AFRICA

- 1 No significant iron-ore reserves in Africa except in South Africa in the Transvaal region.
- 2 Most of the ores from here goes into Chinese blast-furnace.

IRON-ORE IN AUSTRALIA

From Pilbara region most of the ore goes to Chinese and Japanese blast-furnaces.

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